Resumen: Estadísticos, Intervalos de Confianza, Contrastes de Hipótesis y Regresión Lineal

1. Estadísticos y sus Distribuciones

Cuadro 1: Distribuciones de estadísticos usuales

Parámetro	**Población(es)**	**Condiciones**	**Estadístico**	**Distribución**
Media μ	$X \sim N(\mu, \sigma)$	σ conocida o $n \geq 30$	$Z = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$N(0,1)$
Media μ	$X \sim N(\mu, \sigma)$	σ desconocida	$T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}}$	t_{n-1}
Varianza σ^2	$X \sim N(\mu, \sigma)$	Población normal	$\chi^2 = \dfrac{(n-1)S^2}{\sigma^2}$	χ^2_{n-1}
Dif. medias $\mu_1 - \mu_2$	$X \sim N(\mu_1, \sigma_1)$, $Y \sim N(\mu_2, \sigma_2)$	σ_1, σ_2 conocidas	$Z = \dfrac{(\bar{X} - \bar{Y})}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$N(0,1)$
Dif. medias $\mu_1 - \mu_2$	$X \sim N(\mu_1, \sigma_1)$, $Y \sim N(\mu_2, \sigma_2)$	$\sigma_1 = \sigma_2$ desconocidas	$T = \dfrac{(\bar{X} - \bar{Y})}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$t_{n_1+n_2-2}$
Dif. medias $\mu_1 - \mu_2$	$X \sim N(\mu_1, \sigma_1)$, $Y \sim N(\mu_2, \sigma_2)$	$\sigma_1 \neq \sigma_2$ desconocidas	$T = \dfrac{(\bar{X} - \bar{Y})}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$	t_ν
Cociente varianzas $\frac{\sigma_1^2}{\sigma_2^2}$	$X \sim N(\mu_1, \sigma_1)$, $Y \sim N(\mu_2, \sigma_2)$	Poblaciones normales	$F = \dfrac{S_1^2}{S_2^2}$	F_{n_1-1,n_2-1}
Proporción p	$X \sim B(1, p)$	$n \geq 30$	$Z = \dfrac{\hat{p} - p}{\sqrt{p(1-p)/n}}$	$N(0,1)$
Dif. proporciones $p_1 - p_2$	$X \sim B(1, p_1)$, $Y \sim B(1, p_2)$	$n_1, n_2 \geq 30$	$Z = \dfrac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$	$N(0,1)$

Fórmulas auxiliares:

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}, \quad \nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}}$$

2. Intervalos de Confianza

Cuadro 2: Intervalos de Confianza al nivel $(1-\alpha)\,\%$

Parámetro	**Condiciones**	**IC Bilateral**
Media μ	σ conocida	$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
Media μ	σ desconocida	$\bar{x} \pm t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}$
Varianza σ^2	Población normal	$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right)$
Dif. medias $\mu_1 - \mu_2$	σ_1, σ_2 conocidas	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
Dif. medias $\mu_1 - \mu_2$	$\sigma_1 = \sigma_2$ desconocidas	$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
Dif. medias $\mu_1 - \mu_2$	$\sigma_1 \neq \sigma_2$ desconocidas	$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\nu} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Cociente varianzas $\frac{\sigma_1^2}{\sigma_2^2}$	Poblaciones normales	$\left(\frac{s_1^2/s_2^2}{F_{\alpha/2,n_1-1,n_2-1}}, \frac{s_1^2/s_2^2}{F_{1-\alpha/2,n_1-1,n_2-1}}\right)$
Proporción p	$n\hat{p} \geq 5$, $n(1-\hat{p}) \geq 5$	$\hat{p} \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Dif. proporciones $p_1 - p_2$	Condiciones binomiales	$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

3. Contrastes de Hipótesis

3.1. Una Población

Cuadro 3: Contrastes para una población

Parámetro	**Hipótesis**	**Estadístico**	**Región Crítica**
Media μ	$H_0: \mu = \mu_0$ vs $H_a: \mu \neq \mu_0$ $H_0: \mu \leq \mu_0$ vs $H_a: \mu > \mu_0$ $H_0: \mu \geq \mu_0$ vs $H_a: \mu < \mu_0$	$Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$	$\lvert Z\rvert > z_{\alpha/2}$ $Z > z_\alpha$ $Z < -z_\alpha$
Media μ	$H_0: \mu = \mu_0$ vs $H_a: \mu \neq \mu_0$ $H_0: \mu \leq \mu_0$ vs $H_a: \mu > \mu_0$ $H_0: \mu \geq \mu_0$ vs $H_a: \mu < \mu_0$	$T = \frac{\bar{X}-\mu_0}{S/\sqrt{n}}$	$\lvert T\rvert > t_{\alpha/2,n-1}$ $T > t_{\alpha,n-1}$ $T < -t_{\alpha,n-1}$
Varianza σ^2	$H_0: \sigma^2 = \sigma_0^2$ vs $H_a: \sigma^2 \neq \sigma_0^2$ $H_0: \sigma^2 \leq \sigma_0^2$ vs $H_a: \sigma^2 > \sigma_0^2$ $H_0: \sigma^2 \geq \sigma_0^2$ vs $H_a: \sigma^2 < \sigma_0^2$	$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}$	$\chi^2 < \chi^2_{1-\alpha/2,n-1}$ o $\chi^2 > \chi^2_{\alpha/2,n-1}$ $\chi^2 > \chi^2_{\alpha,n-1}$ $\chi^2 < \chi^2_{1-\alpha,n-1}$
Proporción p	$H_0: p = p_0$ vs $H_a: p \neq p_0$ $H_0: p \leq p_0$ vs $H_a: p > p_0$ $H_0: p \geq p_0$ vs $H_a: p < p_0$	$Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$	$\lvert Z\rvert > z_{\alpha/2}$ $Z > z_\alpha$ $Z < -z_\alpha$

3.2. Dos Poblaciones

Cuadro 4: Contrastes para dos poblaciones

Parámetro	**Hipótesis**	**Estadístico**	**Región Crítica**
Dif. medias	$H_0 : \mu_1 = \mu_2$ vs $H_a : \mu_1 \neq \mu_2$ $H_0 : \mu_1 \leq \mu_2$ vs $H_a : \mu_1 > \mu_2$ $H_0 : \mu_1 \geq \mu_2$ vs $H_a : \mu_1 < \mu_2$	$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$\lvert Z \rvert > z_{\alpha/2}$ $Z > z_\alpha$ $Z < -z_\alpha$
Dif. medias $\sigma_1 = \sigma_2$ desconocidas	$H_0 : \mu_1 = \mu_2$ vs $H_a : \mu_1 \neq \mu_2$ $H_0 : \mu_1 \leq \mu_2$ vs $H_a : \mu_1 > \mu_2$ $H_0 : \mu_1 \geq \mu_2$ vs $H_a : \mu_1 < \mu_2$	$T = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$\lvert T \rvert > t_{\alpha/2, n_1+n_2-2}$ $T > t_{\alpha, n_1+n_2-2}$ $T < -t_{\alpha, n_1+n_2-2}$
Dif. medias	$H_0 : \mu_1 = \mu_2$ vs $H_a : \mu_1 \neq \mu_2$ $H_0 : \mu_1 \leq \mu_2$ vs $H_a : \mu_1 > \mu_2$ $H_0 : \mu_1 \geq \mu_2$ vs $H_a : \mu_1 < \mu_2$	$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$	$\lvert T \rvert > t_{\alpha/2, \nu}$ $T > t_{\alpha, \nu}$ $T < -t_{\alpha, \nu}$
Cociente varianzas	$H_0 : \sigma_1^2 = \sigma_2^2$ vs $H_a : \sigma_1^2 \neq \sigma_2^2$ $H_0 : \sigma_1^2 \leq \sigma_2^2$ vs $H_a : \sigma_1^2 > \sigma_2^2$ $H_0 : \sigma_1^2 \geq \sigma_2^2$ vs $H_a : \sigma_1^2 < \sigma_2^2$	$F = \frac{S_1^2}{S_2^2}$	$F < F_{1-\alpha/2, n_1-1, n_2-1}$ o $F > F_{\alpha/2, n_1-1, n_2-1}$ $F > F_{\alpha, n_1-1, n_2-1}$ $F < F_{1-\alpha, n_1-1, n_2-1}$
Dif. proporciones	$H_0 : p_1 = p_2$ vs $H_a : p_1 \neq p_2$ $H_0 : p_1 \leq p_2$ vs $H_a : p_1 > p_2$ $H_0 : p_1 \geq p_2$ vs $H_a : p_1 < p_2$	$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$	$\lvert Z \rvert > z_{\alpha/2}$ $Z > z_\alpha$ $Z < -z_\alpha$

Nota:

- Para proporciones: $\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$ (proporción combinada)

4. Errores en Contraste de Hipótesis

Cuadro 5: Tipos de errores

	H_0 **verdadera**	H_0 **falsa**
No rechazar H_0	Decisión correcta	Error Tipo II (β)
Rechazar H_0	Error Tipo I (α)	Decisión correcta

- $\alpha = P(\text{Error Tipo I}) = P(\text{Rechazar } H_0 | H_0 \text{ verdadera})$
- $\beta = P(\text{Error Tipo II}) = P(\text{No rechazar } H_0 | H_0 \text{ falsa})$
- Potencia $= 1 - \beta = P(\text{Rechazar } H_0 | H_0 \text{ falsa})$

5. Valor p

Definición: Probabilidad de obtener un resultado al menos tan extremo como el observado, suponiendo H_0 verdadera.

Interpretación:

- Si valor $p < \alpha$: Rechazar H_0
- Si valor $p \geq \alpha$: No rechazar H_0

Cálculo:

- **Bilateral**: valor $p = 2 \cdot P(Z \geq |z|)$
- **Unilateral derecho**: valor $p = P(Z \geq z)$
- **Unilateral izquierdo**: valor $p = P(Z \leq z)$

6. Ajuste de Bondad (Prueba Chi-cuadrado)

Cuadro 6: Prueba Chi-cuadrado de bondad de ajuste

Prueba	**Hipótesis**	**Estadístico**	**Distribución**
Chi-cuadrado	H_0: Los datos siguen la distribución especificada	$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$	χ^2_{k-1-m}

Región crítica: $\chi^2 > \chi^2_{\alpha,k-1-m}$

7. ANOVA (Análisis de la Varianza)

Cuadro 7: Tabla ANOVA

Fuente	**Suma de Cuadrados**	**gl**	**Cuadrado Medio**	**F**
Entre grupos	$SCE = \sum n_i(\bar{x}_i - \bar{x})^2$	$k-1$	$CME = \frac{SCE}{k-1}$	$F = \frac{CME}{CMR}$
Dentro grupos	$SCR = \sum\sum(x_{ij} - \bar{x}_i)^2$	$n-k$	$CMR = \frac{SCR}{n-k}$	
Total	$SCT = \sum\sum(x_{ij} - \bar{x})^2$	$n-1$		

Contraste:

- Estadístico: $F = \frac{CME}{CMR} \sim F_{k-1,n-k}$
- Se rechaza H_0 si $F > F_{\alpha,k-1,n-k}$

8. Regresión Lineal

8.1. Regresión lineal simple

Modelo: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $\epsilon_i \sim N(0, \sigma^2)$
Estimadores:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Anexo: Interpolación Lineal

Fórmula: Para interpolar entre (x_0, y_0) y (x_1, y_1), con $x_0 < x < x_1$:

$$y = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0)$$

7. Distribución χ^2 Ji-Cuadrada

$$Y \sim \chi^2_n$$

siendo n los grados de libertad.

$$p = P(Y \leq y) = \int_0^y f_Y(u)du = 1 - \alpha$$

donde, para $u \geq 0$,

$$f_Y(u) = \frac{1}{2^{n/2}\Gamma(n/2)} u^{n/2-1} e^{-u/2}$$

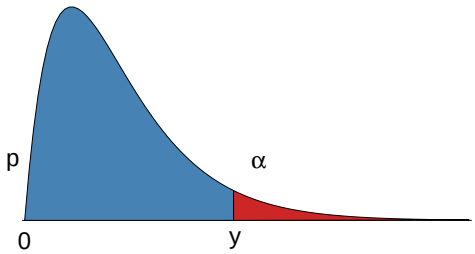


Tabla 7. Valores críticos $\chi^2_{(\alpha;n)}$ de la distribución χ^2_n Ji-Cuadrada.

	p									
	0.005	0.01	0.025	0.05	0.1	0.90	0.95	0.975	0.99	0.995
	α									
n	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
31	14.458	15.655	17.539	19.281	21.434	41.422	44.985	48.232	52.191	55.003
32	15.134	16.362	18.291	20.072	22.271	42.585	46.194	49.480	53.486	56.328
33	15.815	17.074	19.047	20.867	23.110	43.745	47.400	50.725	54.776	57.648
34	16.501	17.789	19.806	21.664	23.952	44.903	48.602	51.966	56.061	58.964
35	17.192	18.509	20.569	22.465	24.797	46.059	49.802	53.203	57.342	60.275
36	17.887	19.233	21.336	23.269	25.643	47.212	50.998	54.437	58.619	61.581
37	18.586	19.960	22.106	24.075	26.492	48.363	52.192	55.668	59.893	62.883
38	19.289	20.691	22.878	24.884	27.343	49.513	53.384	56.896	61.162	64.181
39	19.996	21.426	23.654	25.695	28.196	50.660	54.572	58.120	62.428	65.476
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
75	47.206	49.475	52.942	56.054	59.795	91.061	96.217	100.839	106.393	110.286
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

8. Distribución t de Student

$$T \sim t_n$$

siendo n los grados de libertad.

$$p = P(T \leq t) = \int_{-\infty}^{t} f_T(u)du = 1 - \alpha$$

donde, para $-\infty < u < \infty$,

$$f_T(u) = \frac{1}{\sqrt{n\pi}} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{u^2}{n}\right)^{-\frac{n+1}{2}}$$

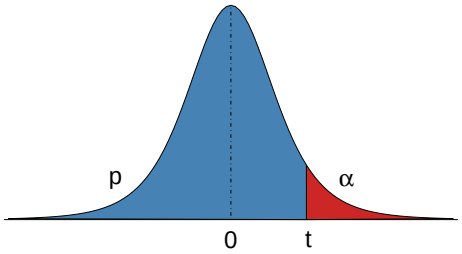


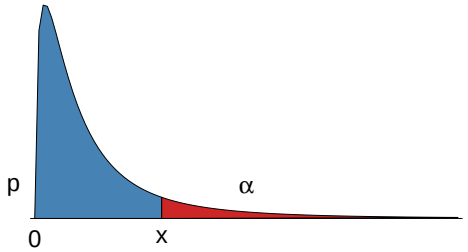
Tabla 8. Valores críticos $t_{(\alpha;n)}$ de la distribución t de Student.

						p				
	0.75	0.80	0.90	0.95	0.975	0.99	0.995	0.999	0.9995	0.9999
						α				
n	0.25	0.20	0.10	0.05	0.025	0.01	0.005	0.001	0.0005	0.0001
1	1.000	1.376	3.078	6.314	12.706	31.821	63.657	318.309	636.619	3183.099
2	0.816	1.061	1.886	2.920	4.303	6.965	9.925	22.327	31.599	70.700
3	0.765	0.978	1.638	2.353	3.182	4.541	5.841	10.215	12.924	22.204
4	0.741	0.941	1.533	2.132	2.776	3.747	4.604	7.173	8.610	13.034
5	0.727	0.920	1.476	2.015	2.571	3.365	4.032	5.893	6.869	9.678
6	0.718	0.906	1.440	1.943	2.447	3.143	3.707	5.208	5.959	8.025
7	0.711	0.896	1.415	1.895	2.365	2.998	3.499	4.785	5.408	7.063
8	0.706	0.889	1.397	1.860	2.306	2.896	3.355	4.501	5.041	6.442
9	0.703	0.883	1.383	1.833	2.262	2.821	3.250	4.297	4.781	6.010
10	0.700	0.879	1.372	1.812	2.228	2.764	3.169	4.144	4.587	5.694
11	0.697	0.876	1.363	1.796	2.201	2.718	3.106	4.025	4.437	5.453
12	0.695	0.873	1.356	1.782	2.179	2.681	3.055	3.930	4.318	5.263
13	0.694	0.870	1.350	1.771	2.160	2.650	3.012	3.852	4.221	5.111
14	0.692	0.868	1.345	1.761	2.145	2.624	2.977	3.787	4.140	4.985
15	0.691	0.866	1.341	1.753	2.131	2.602	2.947	3.733	4.073	4.880
16	0.690	0.865	1.337	1.746	2.120	2.583	2.921	3.686	4.015	4.791
17	0.689	0.863	1.333	1.740	2.110	2.567	2.898	3.646	3.965	4.714
18	0.688	0.862	1.330	1.734	2.101	2.552	2.878	3.610	3.922	4.648
19	0.688	0.861	1.328	1.729	2.093	2.539	2.861	3.579	3.883	4.590
20	0.687	0.860	1.325	1.725	2.086	2.528	2.845	3.552	3.850	4.539
21	0.686	0.859	1.323	1.721	2.080	2.518	2.831	3.527	3.819	4.493
22	0.686	0.858	1.321	1.717	2.074	2.508	2.819	3.505	3.792	4.452
23	0.685	0.858	1.319	1.714	2.069	2.500	2.807	3.485	3.768	4.415
24	0.685	0.857	1.318	1.711	2.064	2.492	2.797	3.467	3.745	4.382
25	0.684	0.856	1.316	1.708	2.060	2.485	2.787	3.450	3.725	4.352
26	0.684	0.856	1.315	1.706	2.056	2.479	2.779	3.435	3.707	4.324
27	0.684	0.855	1.314	1.703	2.052	2.473	2.771	3.421	3.690	4.299
28	0.683	0.855	1.313	1.701	2.048	2.467	2.763	3.408	3.674	4.275
29	0.683	0.854	1.311	1.699	2.045	2.462	2.756	3.396	3.659	4.254
30	0.683	0.854	1.310	1.697	2.042	2.457	2.750	3.385	3.646	4.234
40	0.681	0.851	1.303	1.684	2.021	2.423	2.704	3.307	3.551	4.094
50	0.679	0.849	1.299	1.676	2.009	2.403	2.678	3.261	3.496	4.014
75	0.678	0.846	1.293	1.665	1.992	2.377	2.643	3.202	3.425	3.911
100	0.677	0.845	1.290	1.660	1.984	2.364	2.626	3.174	3.390	3.862
125	0.676	0.845	1.288	1.657	1.979	2.357	2.616	3.157	3.370	3.832
∞	0.674	0.842	1.282	1.645	1.960	2.326	2.576	3.090	3.291	3.719

9. Distribución F

$$X \sim F_{n_1,n_2}$$

con n_1 y n_2 los grados de libertad (del numerador y denominador, respectivamente).



$$p = P(X \leq x) = \int_0^x f_X(u)du = 1 - \alpha$$

donde, para $u > 0$,

$$f_X(u) = \frac{\Gamma\big((n_1+n_2)/2\big)}{\Gamma(n_1/2)\Gamma(n_2/2)} \left(\frac{n_1}{n_2}\right)^{n_1/2} \frac{u^{n_1/2-1}}{[1+(n_1/n_2)u]^{(n_1+n_2)/2}}$$

Nota: Si $X \sim F_{n_1,n_2}$, entonces,

$$p = P(X \leq F_{(1-\alpha;\, n_1,n_2)}) = P\left(X \leq \frac{1}{F_{(\alpha;\, n_2,n_1)}}\right) = 1 - \alpha$$

Tabla 9B. Valores críticos $F_{(\alpha;n_1,n_2)}$ de la distribución F.

$p = 0.95$ $\alpha = 0.05$

n_2 \ n_1	1	2	3	4	5	6	7	8	9	10	12	15	20	25	50	75	100	∞
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.95	248.01	249.26	251.77	252.62	253.04	254.30
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.46	19.48	19.48	19.49	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.63	8.58	8.56	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.70	5.68	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.52	4.44	4.42	4.41	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.83	3.75	3.73	3.71	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.40	3.32	3.29	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.11	3.02	2.99	2.97	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.89	2.80	2.77	2.76	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.73	2.64	2.60	2.59	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.60	2.51	2.47	2.46	2.41
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.50	2.40	2.37	2.35	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.41	2.31	2.28	2.26	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.34	2.24	2.21	2.19	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.28	2.18	2.14	2.12	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.23	2.12	2.09	2.07	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.18	2.08	2.04	2.02	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.14	2.04	2.00	1.98	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.00	1.96	1.94	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.07	1.97	1.93	1.91	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	1.94	1.90	1.88	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.02	1.91	1.87	1.85	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.00	1.88	1.84	1.82	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.97	1.86	1.82	1.80	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.84	1.80	1.78	1.71
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.07	1.99	1.94	1.82	1.78	1.76	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.06	1.97	1.92	1.81	1.76	1.74	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.04	1.96	1.91	1.79	1.75	1.73	1.65
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.03	1.94	1.89	1.77	1.73	1.71	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.88	1.76	1.72	1.70	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.78	1.66	1.61	1.59	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.69	1.56	1.51	1.48	1.39
80	3.96	3.11	2.72	2.49	2.33	2.21	2.13	2.06	2.00	1.95	1.88	1.79	1.70	1.64	1.51	1.45	1.43	1.33
100	3.94	3.09	2.70	2.46	2.31	2.19	2.10	2.03	1.97	1.93	1.85	1.77	1.68	1.62	1.48	1.42	1.39	1.28
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.60	1.46	1.40	1.37	1.26
∞	3.84	3.00	2.61	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	1.57	1.51	1.35	1.28	1.25	1.03

Tabla 9C. Valores críticos $F_{(\alpha;n_1,n_2)}$ de la distribución F.

$p = 0.975$ $\alpha = 0.025$

n_2 \ n_1	1	2	3	4	5	6	7	8	9	10	12	15	20	25	50	75	100	∞
1	648.	799.	864.	900.	922.	937.	948.	957.	963.	969.	977.	985.	993.	998.	1008.	1011.	1013.	1018.
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	39.46	39.48	39.48	39.49	39.50
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	14.12	14.01	13.97	13.96	13.90
4	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84	8.75	8.66	8.56	8.50	8.38	8.34	8.32	8.26
5	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62	6.52	6.43	6.33	6.27	6.14	6.10	6.08	6.02
6	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46	5.37	5.27	5.17	5.11	4.98	4.94	4.92	4.85
7	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76	4.67	4.57	4.47	4.40	4.28	4.23	4.21	4.14
8	7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30	4.20	4.10	4.00	3.94	3.81	3.76	3.74	3.67
9	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96	3.87	3.77	3.67	3.60	3.47	3.43	3.40	3.33
10	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72	3.62	3.52	3.42	3.35	3.22	3.18	3.15	3.08
11	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.66	3.59	3.53	3.43	3.33	3.23	3.16	3.03	2.98	2.96	2.88
12	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.51	3.44	3.37	3.28	3.18	3.07	3.01	2.87	2.82	2.80	2.73
13	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.39	3.31	3.25	3.15	3.05	2.95	2.88	2.74	2.70	2.67	2.60
14	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.29	3.21	3.15	3.05	2.95	2.84	2.78	2.64	2.59	2.56	2.49
15	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.20	3.12	3.06	2.96	2.86	2.76	2.69	2.55	2.50	2.47	2.40
16	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.12	3.05	2.99	2.89	2.79	2.68	2.61	2.47	2.42	2.40	2.32
17	6.04	4.62	4.01	3.66	3.44	3.28	3.16	3.06	2.98	2.92	2.82	2.72	2.62	2.55	2.41	2.35	2.33	2.25
18	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01	2.93	2.87	2.77	2.67	2.56	2.49	2.35	2.30	2.27	2.19
19	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96	2.88	2.82	2.72	2.62	2.51	2.44	2.30	2.24	2.22	2.13
20	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91	2.84	2.77	2.68	2.57	2.46	2.40	2.25	2.20	2.17	2.09
21	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87	2.80	2.73	2.64	2.53	2.42	2.36	2.21	2.16	2.13	2.04
22	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84	2.76	2.70	2.60	2.50	2.39	2.32	2.17	2.12	2.09	2.00
23	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81	2.73	2.67	2.57	2.47	2.36	2.29	2.14	2.08	2.06	1.97
24	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78	2.70	2.64	2.54	2.44	2.33	2.26	2.11	2.05	2.02	1.94
25	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75	2.68	2.61	2.51	2.41	2.30	2.23	2.08	2.02	2.00	1.91
26	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73	2.65	2.59	2.49	2.39	2.28	2.21	2.05	2.00	1.97	1.88
27	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71	2.63	2.57	2.47	2.36	2.25	2.18	2.03	1.97	1.94	1.85
28	5.61	4.22	3.63	3.29	3.06	2.90	2.78	2.69	2.61	2.55	2.45	2.34	2.23	2.16	2.01	1.95	1.92	1.83
29	5.59	4.20	3.61	3.27	3.04	2.88	2.76	2.67	2.59	2.53	2.43	2.32	2.21	2.14	1.99	1.93	1.90	1.81
30	5.57	4.18	3.59	3.25	3.03	2.87	2.75	2.65	2.57	2.51	2.41	2.31	2.20	2.12	1.97	1.91	1.88	1.79
40	5.42	4.05	3.46	3.13	2.90	2.74	2.62	2.53	2.45	2.39	2.29	2.18	2.07	1.99	1.83	1.77	1.74	1.64
60	5.29	3.93	3.34	3.01	2.79	2.63	2.51	2.41	2.33	2.27	2.17	2.06	1.94	1.87	1.70	1.63	1.60	1.48
80	5.22	3.86	3.28	2.95	2.73	2.57	2.45	2.35	2.28	2.21	2.11	2.00	1.88	1.81	1.63	1.56	1.53	1.40
100	5.18	3.83	3.25	2.92	2.70	2.54	2.42	2.32	2.24	2.18	2.08	1.97	1.85	1.77	1.59	1.52	1.48	1.35
120	5.15	3.80	3.23	2.89	2.67	2.52	2.39	2.30	2.22	2.16	2.05	1.94	1.82	1.75	1.56	1.49	1.45	1.31
∞	5.03	3.69	3.12	2.79	2.57	2.41	2.29	2.19	2.11	2.05	1.95	1.83	1.71	1.63	1.43	1.35	1.30	1.04

Tabla 9E. Valores críticos $F_{(\alpha;n_1,n_2)}$ de la distribución F.

$p = 0.995$ $\alpha = 0.005$

n_2 \ n_1	1	2	3	4	5	6	7	8	9	10	12	15	20	25	50	75	100	∞
1	16211.	19999.	21615.	22500.	23056.	23437.	23715.	23925.	24091.	24224.	24426.	24630.	24836.	24960.	25211.	25295.	25337.	25463.
2	198.50	199.00	199.17	199.25	199.30	199.33	199.36	199.37	199.39	199.40	199.42	199.43	199.45	199.46	199.48	199.49	199.49	199.50
3	55.55	49.80	47.47	46.19	45.39	44.84	44.43	44.13	43.88	43.69	43.39	43.08	42.78	42.59	42.21	42.09	42.02	41.83
4	31.33	26.28	24.26	23.15	22.46	21.97	21.62	21.35	21.14	20.97	20.70	20.44	20.17	20.00	19.67	19.55	19.50	19.33
5	22.78	18.31	16.53	15.56	14.94	14.51	14.20	13.96	13.77	13.62	13.38	13.15	12.90	12.76	12.45	12.35	12.30	12.15
6	18.63	14.54	12.92	12.03	11.46	11.07	10.79	10.57	10.39	10.25	10.03	9.81	9.59	9.45	9.17	9.07	9.03	8.88
7	16.24	12.40	10.88	10.05	9.52	9.16	8.89	8.68	8.51	8.38	8.18	7.97	7.75	7.62	7.35	7.26	7.22	7.08
8	14.69	11.04	9.60	8.81	8.30	7.95	7.69	7.50	7.34	7.21	7.01	6.81	6.61	6.48	6.22	6.13	6.09	5.95
9	13.61	10.11	8.72	7.96	7.47	7.13	6.88	6.69	6.54	6.42	6.23	6.03	5.83	5.71	5.45	5.37	5.32	5.19
10	12.83	9.43	8.08	7.34	6.87	6.54	6.30	6.12	5.97	5.85	5.66	5.47	5.27	5.15	4.90	4.82	4.77	4.64
11	12.23	8.91	7.60	6.88	6.42	6.10	5.86	5.68	5.54	5.42	5.24	5.05	4.86	4.74	4.49	4.40	4.36	4.23
12	11.75	8.51	7.23	6.52	6.07	5.76	5.52	5.35	5.20	5.09	4.91	4.72	4.53	4.41	4.17	4.08	4.04	3.91
13	11.37	8.19	6.93	6.23	5.79	5.48	5.25	5.08	4.94	4.82	4.64	4.46	4.27	4.15	3.91	3.82	3.78	3.65
14	11.06	7.92	6.68	6.00	5.56	5.26	5.03	4.86	4.72	4.60	4.43	4.25	4.06	3.94	3.70	3.61	3.57	3.44
15	10.80	7.70	6.48	5.80	5.37	5.07	4.85	4.67	4.54	4.42	4.25	4.07	3.88	3.77	3.52	3.44	3.39	3.26
16	10.58	7.51	6.30	5.64	5.21	4.91	4.69	4.52	4.38	4.27	4.10	3.92	3.73	3.62	3.37	3.29	3.25	3.11
17	10.38	7.35	6.16	5.50	5.07	4.78	4.56	4.39	4.25	4.14	3.97	3.79	3.61	3.49	3.25	3.16	3.12	2.99
18	10.22	7.21	6.03	5.37	4.96	4.66	4.44	4.28	4.14	4.03	3.86	3.68	3.50	3.38	3.14	3.05	3.01	2.87
19	10.07	7.09	5.92	5.27	4.85	4.56	4.34	4.18	4.04	3.93	3.76	3.59	3.40	3.29	3.04	2.96	2.91	2.78
20	9.94	6.99	5.82	5.17	4.76	4.47	4.26	4.09	3.96	3.85	3.68	3.50	3.32	3.20	2.96	2.87	2.83	2.69
21	9.83	6.89	5.73	5.09	4.68	4.39	4.18	4.01	3.88	3.77	3.60	3.43	3.24	3.13	2.88	2.80	2.75	2.62
22	9.73	6.81	5.65	5.02	4.61	4.32	4.11	3.94	3.81	3.70	3.54	3.36	3.18	3.06	2.82	2.73	2.69	2.55
23	9.63	6.73	5.58	4.95	4.54	4.26	4.05	3.88	3.75	3.64	3.47	3.30	3.12	3.00	2.76	2.67	2.62	2.49
24	9.55	6.66	5.52	4.89	4.49	4.20	3.99	3.83	3.69	3.59	3.42	3.25	3.06	2.95	2.70	2.61	2.57	2.43
25	9.48	6.60	5.46	4.84	4.43	4.15	3.94	3.78	3.64	3.54	3.37	3.20	3.01	2.90	2.65	2.56	2.52	2.38
26	9.41	6.54	5.41	4.79	4.38	4.10	3.89	3.73	3.60	3.49	3.33	3.15	2.97	2.85	2.61	2.52	2.47	2.33
27	9.34	6.49	5.36	4.74	4.34	4.06	3.85	3.69	3.56	3.45	3.28	3.11	2.93	2.81	2.57	2.48	2.43	2.29
28	9.28	6.44	5.32	4.70	4.30	4.02	3.81	3.65	3.52	3.41	3.25	3.07	2.89	2.77	2.53	2.44	2.39	2.25
29	9.23	6.40	5.28	4.66	4.26	3.98	3.77	3.61	3.48	3.38	3.21	3.04	2.86	2.74	2.49	2.40	2.36	2.21
30	9.18	6.35	5.24	4.62	4.23	3.95	3.74	3.58	3.45	3.34	3.18	3.01	2.82	2.71	2.46	2.37	2.32	2.18
40	8.83	6.07	4.98	4.37	3.99	3.71	3.51	3.35	3.22	3.12	2.95	2.78	2.60	2.48	2.23	2.14	2.09	1.93
60	8.49	5.79	4.73	4.14	3.76	3.49	3.29	3.13	3.01	2.90	2.74	2.57	2.39	2.27	2.01	1.91	1.86	1.69
80	8.33	5.67	4.61	4.03	3.65	3.39	3.19	3.03	2.91	2.80	2.64	2.47	2.29	2.17	1.90	1.80	1.75	1.57
100	8.24	5.59	4.54	3.96	3.59	3.33	3.13	2.97	2.85	2.74	2.58	2.41	2.23	2.11	1.84	1.74	1.68	1.49
120	8.18	5.54	4.50	3.92	3.55	3.28	3.09	2.93	2.81	2.71	2.54	2.37	2.19	2.07	1.80	1.69	1.64	1.43
∞	7.88	5.30	4.28	3.72	3.35	3.09	2.90	2.75	2.62	2.52	2.36	2.19	2.00	1.88	1.59	1.47	1.40	1.05